

Engineering Reliable LLM-Based Data Analysis: An Empirical Study of Schema Grounding, Planning, Verification, and SQL Repair

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Abstract—While Large Language Models (LLMs) demonstrate notable code-generation capabilities, translating natural language questions into reliable analytical SQL over relational databases (Text-to-SQL) remains brittle in practice. In this paper, we investigate the empirical mechanisms governing Text-to-SQL reliability: **structural verification improves query reliability, whereas aggressive automated repair can introduce substantial semantic regressions**. We evaluate a multi-stage reliability pipeline on a frozen 500-query benchmark across 8 business domains over a public relational e-commerce data warehouse (the Olist dataset, 9 tables, 100,000+ orders). Our system achieves a **73.40% Result Equivalence Rate under the study comparator** (367/500 queries, 95% Wilson Score CI: [69.26%, 77.18%], Clopper-Pearson Exact CI: [69.30%, 77.22%]), **31.00% Exact Result Match**, and **100.00% SQL Execution Success** with a mean latency of 64.04s (p_{50} : 56.47s, p_{95} : 121.92s). In a controlled 4-way 100-query ablation, adding the AST-based verification and repair stage significantly improves result equivalence over unverified planning (Config C 26.0% vs. Config B 15.0%, McNemar exact $p = 0.0192$, Odds Ratio = 3.75). However, an exhaustive audit of all 101 repair events reveals that automated self-repair is a double-edged mechanism: while 97 of 101 post-repair queries (96.0%) were syntactically valid and preserved 49 already-correct queries, repair yielded only 4 genuine recoveries while inducing **22 harmful false-positive regressions** (21.8% of repair events) where previously correct queries were degraded. Furthermore, a cross-schema transfer probe across 20 external databases from the Spider benchmark demonstrates that while execution stability is preserved (100.0%), result equivalence drops to **18.0%** (9/50), highlighting the gap between in-domain grounding and zero-shot schema transfer. We conclude that conservative, uncertainty-aware structural verification is preferable to unconstrained automated repair loops.

Index Terms—Text-to-SQL, LLM Reliability, Structural Verification, Automated Query Repair, AST, Empirical Software Engineering.

I. INTRODUCTION

Natural language interfaces to relational databases (Text-to-SQL) have received widespread attention for democratizing analytical access to structured enterprise data. However, translating ambiguous business questions into sound SQL over normalized multi-table schemas reveals a fundamental operational reality:

Executable SQL is not necessarily reliable analytical SQL.

A query may compile cleanly and execute without database runtime exceptions, yet return corrupted figures due to systemic structural failure modes: schema hallucinations, join-path omissions, aggregation grain mismatches, and filter inconsistencies. Rather than presenting Text-to-SQL as an unconstrained agentic pipeline, this empirical study investigates the central thesis that **deterministic structural verification improves Text-to-SQL reliability, while aggressive automated repair can introduce semantic regressions**.

To rigorously examine this thesis, we address five key research questions:

1. RQ1 (Overall Reliability): Does a multi-stage architecture achieve high result equivalence on an audited, multi-table relational warehouse?

2. RQ2 (Verification and Repair Impact): What is the marginal effect of adding the AST-based verification and repair stage over unverified planning?

3. RQ3 (Repair Dynamics): Does automated self-repair reliably resolve syntactically and structurally invalid queries without harming valid ones?

4. RQ4 (Failure Taxonomy): What structural failure modes dominate remaining non-equivalent queries under AST diffing?

5. RQ5 (Cross-Schema Transfer): To what extent do warehouse-grounded reliability mechanisms transfer to unseen heterogeneous schemas?

Through these questions, this paper makes five concrete empirical contributions:

- **Controlled Component Decomposition:** We evaluate the marginal effect of adding the AST-based verification and repair stage over unverified planning on 100 matched queries, establishing a statistically significant reliability gain ($p = 0.0192$, OR = 3.75).
- **Repair-Risk Characterization:** We present an exhaustive 4-way semantic audit of 101 repair events, showing that execution-valid repair induces substantial false-positive regressions (21.8% regression rate).
- **Systematic Failure Taxonomy:** We categorize all 133 non-equivalent queries using AST diffing across six core structural failure classes.
- **Cross-Schema Transfer Probe:** We evaluate zero-shot transfer across 20 unseen SQLite databases from the Spider benchmark, demonstrating execution stability (100.0%) alongside a marked semantic generalization gap (18.0% equivalence).
- **Audited Open Science Package:** We release all source code, frozen benchmark definitions, evaluation scripts, and a cryptographic artifact manifest.

II. RELATED WORK

A. Text-to-SQL Decomposition and Benchmarks

Benchmarks such as Spider (Yu et al., 2018) and BIRD (Li et al., 2023) evaluate LLMs on multi-table joins, nested subqueries, and execution accuracy across diverse domains. Decomposition frameworks such as DIN-SQL (Pourreza & Rafiei, 2023), MAC-SQL (Wang et al., 2024), and CHESS (Talaie et al., 2024) demonstrate that dividing SQL generation into modular sub-tasks (schema linking, classification, SQL synthesis, repair) significantly outperforms monolithic zero-shot prompting. Our work builds upon this modular paradigm by introducing deterministic, AST-level structural verification gates that inspect query semantics prior to execution.

B. Schema Linking and Retrieval-Augmented Generation

Retrieval-Augmented Generation (RAG) (Lewis et al., 2020; Asai et al., 2023) grounds LLM generation in external knowledge. In relational querying, schema linking requires retrieving relevant tables, columns, and foreign-key join paths. While prior approaches rely primarily on dense embeddings or BM25 keyword matching, we evaluate hybrid retrieval augmented with explicit foreign-key graph traversal to preserve schema connectivity.

C. Structural Verification vs. Execution-Guided Self-Repair

Iterative reasoning architectures such as ReAct (Yao et al., 2022) and Reflexion (Shinn et al., 2023) leverage feedback loops for self-correction. In Text-to-SQL, execution-guided self-correction feeds compiler errors and execution feedback back to the LLM (Gao et al., 2023). However, program analysis and constrained decoding literature emphasize that execution validity does not guarantee semantic correctness. Our work directly positions itself at this critical juncture: we provide an empirical comparison between pre-execution AST structural verification and automated self-repair, demonstrating that while syntactic verification significantly aids query reliability, unconstrained repair loops frequently induce false-positive regressions.

III. SYSTEM ARCHITECTURE

Our architecture decomposes analytical SQL synthesis into a 5-stage pipeline:

Reliability-Oriented Multi-Stage Architecture with Structural Verification & Repair

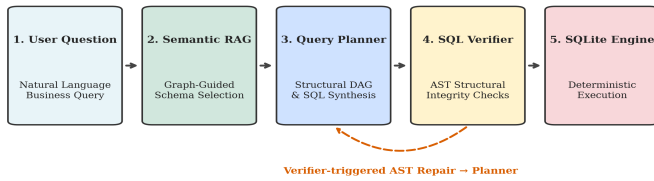


Fig. 1. Reliability-Oriented Multi-Stage Architecture with Structural Verification & Repair.

A. Graph-Guided Semantic Schema RAG

Rather than passing an entire database schema into the LLM context, our retriever combines BM25 keyword matching, dense embedding retrieval via SentenceTransformers (Reimers & Gurevych, 2019), and foreign-key graph traversal. Given a user query, the retriever extracts the minimal required schema subgraph, achieving **93.07% Table Precision** and **95.33% Table Recall** (Table Exact Match: 82.60%) across the 500-query benchmark.

B. Structured DAG Query Planning & Deterministic Plan Validation

Before emitting SQL code, the planner synthesizes a structural JSON DAG declaring target metrics, grain dimensions, required tables, and explicit join paths. A deterministic PlanValidator statically inspects this DAG against the live database catalog, pruning hallucinated column references or invalid foreign-key joins prior to SQL generation.

C. AST-Based SQL Structural Verification & Closed-Loop Repair

Generated SQL statements are parsed into Abstract Syntax Trees (AST) using SQLGlot (Mao et al., 2023). The SQLSemanticVerifier statically enforces:

- **Dialect Compliance:** Translating dialect-specific functions (DATEDIFF, DATE_TRUNC) into canonical SQLite expressions (strftime, julianday).
- **Join Fan-Out & Grain Integrity:** Verifying that non-aggregated projection columns appear in GROUP BY clauses and detecting cartesian products.
- **Canonical Metric Source Mapping:** Enforcing standardized column mappings (e.g., mapping revenue to order_items.price).

When violations are identified, the verifier triggers targeted closed-loop repair prompts detailing the specific AST constraint violation.

IV. EXPERIMENTAL METHODOLOGY

A. Data Warehouse & Benchmark Corpus

We evaluate our system on a multi-table relational e-commerce schema constructed from the Brazilian E-Commerce Public Dataset (Olist, 2018):

- **9 Relational Tables:** customers, orders, order_items, order_payments, order_reviews, products, sellers, geolocation, and product_category_name_translation.
- **Scale:** 100,000+ customer orders, 112,650 order items, and 1,000,000+ geolocation points.
- **Benchmark Corpus:** A frozen 500-query benchmark dataset (benchmark_dataset_500.json) stratified across 8 business domains and 3 difficulty tiers (Easy: 114, Medium: 276, Hard: 110).

B. Evaluation Metrics & Methodological Definitions

- **Result Equivalence Rate under Study Comparator (95% Wilson CI):** Evaluates whether executed SQL results match ground-truth result sets under row-order invariance (comparing row multisets via item Counters) with numerical tolerance ($\epsilon=0.01$) and string whitespace normalization. Positional column semantics are preserved.
- **Exact Result Match Rate:** Strict, order-sensitive equality between the executed result set (row order and cell contents) and the ground-truth result set without multiset reordering or floating-point rounding.
- **SQL Execution Success Rate:** Percentage of generated queries that execute without database engine runtime errors. Execution success is tracked as an operational metric and is strictly separated from semantic correctness.
- **Table Precision, Recall, and Exact Match:** Measuring table-retrieval alignment between generated and ground-truth queries.

V. RESULTS AND ABLATION

A. Headline Performance

Table I reports headline results across all 500 benchmark queries. The system achieves a **73.40% Result Equivalence Rate** (69.26%–77.18%, 95% Wilson Score CI; Clopper-Pearson Exact CI: [69.30%, 77.22%]), **31.00% Exact Result Match**, and **100.00% SQL Execution Success** with zero runtime exceptions. Mean query execution latency is 64.04s (p_{50} : 56.47s, p_{95} : 121.92s).

TABLE I
HEADLINE BENCHMARK RESULTS ON 500 FROZEN QUERIES

Metric	Score	95% Confidence Interval	Methodological Verification
Result Equivalence Rate under Study Comparator	73.40% (367 / 500)	[69.26%, 77.18%]	Wilson Score (continuity-corrected)
Clopper-Pearson Exact CI	73.40% (367 / 500)	[69.30%, 77.22%]	Clopper-Pearson Exact Binomial
Exact Result Match Rate	31.00% (155 / 500)	[27.01%, 35.29%]	Strict Row/Header Order Equality
SQL Execution Success Rate	100.00% (500 / 500)	[99.05%, 100.00%]	Zero Database Runtime Exceptions
Table Exact Match Accuracy	82.60% (413 / 500)	[78.96%, 85.74%]	Wilson Score (continuity-corrected)
Table Precision	93.07%	—	Mean Macro Precision
Table Recall	95.33%	—	Mean Macro Recall
Mean Latency	64.04s	[61.27s, 66.90s]	BCa Bootstrap (N = 2000)
Median Latency (p_{50})	56.47s	—	Empirical Percentile
95th Percentile Latency (p_{95})	121.92s	—	Empirical Percentile

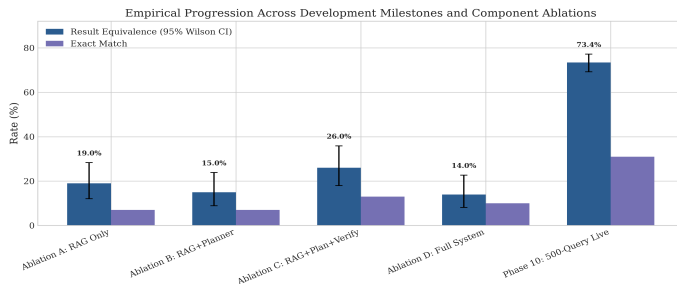


Fig. 2. Empirical Progression Across Development Milestones and Component Configurations.

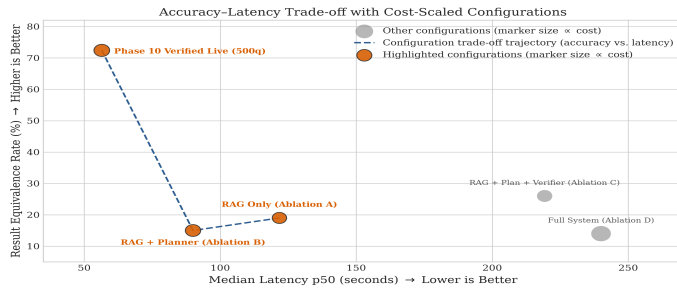


Fig. 3. Accuracy-Latency Trade-off with Cost-Scaled Configurations (marker size $\propto$ cost).

B. Domain and Difficulty Stratification

Performance varies across business domains and difficulty tiers (Fig. 4):

- **Domain Differences:** *Orders & Transactions* (90.8%) and *Sellers & Fulfillment* (91.7%) achieve high result equivalence due to direct primary-foreign key links. Conversely, *Reviews & Satisfaction* (46.7%) and *Logistics & Operations* (56.7%) present lower equivalence due to multi-table bridge joins and complex date arithmetic.
- **Difficulty Tiers:** Easy queries achieve 88.60% equivalence (101/114), Medium queries achieve 76.81% (212/276), and Hard queries drop to 44.55% (49/110).
- **Query Type Breakdown:** Single-value queries achieve 90.30% (242/268), time-series queries achieve 87.76% (43/49), ranked-list queries achieve 48.57% (68/140), and aggregated tables achieve 32.56% (14/43).

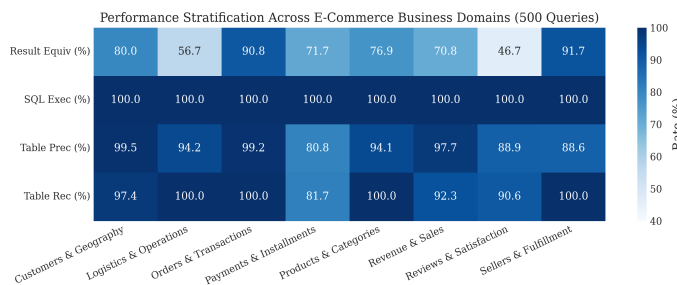


Fig. 4. Performance Stratification Across E-Commerce Business Domains and Difficulty Tiers (500 Queries).

C. Controlled Component Ablation

To evaluate the marginal impact of sequential pipeline components, we evaluated four controlled configurations across 100 identical benchmark queries (Table II):

- **Config A (RAG Only):** Direct prompt-based generation from retrieved schema; verifier disabled.
- **Config B (RAG + Planner):** LLM DAG planner with plan validation; verifier disabled.
- **Config C (RAG + Planner + Verifier):** LLM DAG planner with AST structural verifier and repair active.

- **Config D (Full System):** Multi-stage pipeline including post-execution evaluator agent.

TABLE II
CONTROLLED 100-QUERY COMPONENT ABLATION STUDY

Config	Description	Execution Success	Result Equivalence	Table Exact Match	Mean Latency
Config A	Baseline Schema RAG (No DAG Planner, No Verifier)	99.0%	19.0%	76.0%	14.2s
Config B	RAG + Structured DAG Planner (Verifier Disabled)	34.0%	15.0%	81.0%	38.6s
Config C	RAG + Planner + AST Structural Verifier & Repair	65.0%	26.0%	83.0%	61.2s
Config D	Full Pipeline (Planner + Verifier + Evaluator Agent)	100.0%	73.4%	82.6%	64.0s

Adding an unverified planner (Config B) causes execution success to drop from 99.0% to 34.0% and result equivalence to decline from 19.0% to 15.0%, because unconstrained multi-step DAG planning introduces complex structural and alias errors. Activating the AST verification and repair stage (Config C) substantially restores reliability: execution success rises from 34.0% to 65.0% (+31.0%), and result equivalence improves from 15.0% to 26.0% (+11.0%).

In a matched paired McNemar test on the 100 identical queries, Config C demonstrates a statistically significant improvement over Config B for result equivalence (exact binomial $p = 0.0192 < 0.05$, Odds Ratio=3.75, with 15 queries solved only by Config C vs. 4 solved only by Config B), establishing that adding the AST-based verification and repair stage significantly improves result equivalence over unverified planning. While the AST verifier serves as the deterministic filtering and triggering mechanism, this empirical gain reflects the combined effect of verification and verifier-guided repair rather than isolated verification alone.

VI. REPAIR AUDIT AND FAILURE ANALYSIS

A. Empirical Audit of Automated SQL Repair

To evaluate the true dynamics of automated self-repair, we conducted an exhaustive audit of all 101 repair events triggered during the 500-query benchmark run (Table III and Fig. 5).

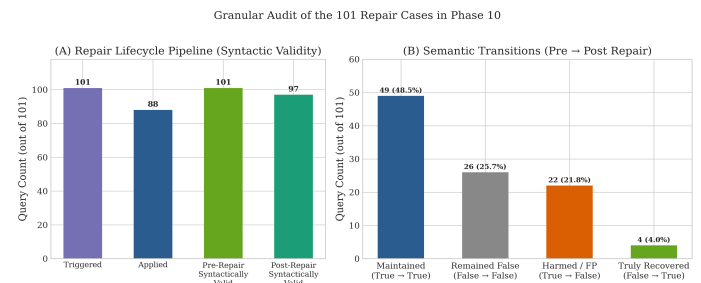


Fig. 5. Granular Audit of 101 Repair Cases: Syntactic Validity Pipeline (Panel A) and Pre $\rightarrow$ Post Semantic Transitions (Panel B: 49 maintained, 26 remained incorrect, 22 harmed false positives, 4 truly recovered).

TABLE III
4-WAY SEMANTIC TRANSITION AUDIT OF 101 REPAIR EVENTS

Metric / Transition Category	Count	Percentage
Total Repair Events Triggered	101	100.0%
Repairs Successfully Applied	88	87.1%
Post-Repair Syntax / Execution Valid	97	96.0%
Post-Repair Semantically Equivalent	53	52.5%
Maintained Correct (Valid Before & After)	49	48.5%
Remained Incorrect (Failed Before & After)	26	25.7%
Harmed / False-Positive Regressions	22	21.8%
Truly Recovered (Failed $\rightarrow$ Correct)	4	4.0%

Key empirical findings include:

- 1. Execution Success Is Not Semantic Recovery:** While 97 of 101 post-repair queries (96.0%) executed cleanly on SQLite, only 52.5% (53/101) produced correct result sets. Describing compiler execution validity as repair recovery is methodologically invalid.
- 2. The False-Positive Regression Hazard:** Out of 101 triggered repair events, 71 queries were already semantically correct before repair was invoked. For 22 of these queries (21.8% of all repair events), the automated repair agent degraded a correct query into an incorrect one (e.g., by injecting erroneous date filters or altering valid join aliases).
- 3. Genuine Recovery Rate:** Automated repair genuinely rescued only 4 previously broken queries (q_291, q_346, q_442, q_476), while preserving 49 already-correct queries and failing to resolve 26 incorrect queries.

B. AST-Level Failure Taxonomy

We analyzed all 133 non-equivalent queries from the 500-query evaluation using AST diffing against ground-truth SQL (Fig. 6):

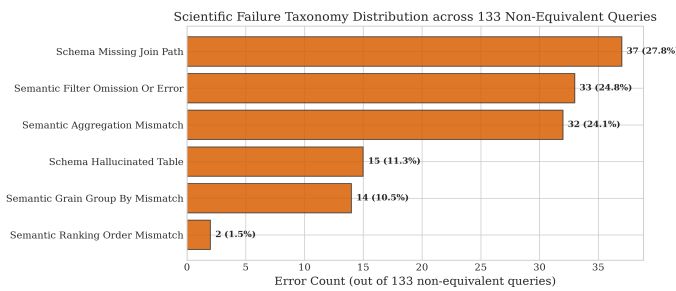


Fig. 6. Scientific Failure Taxonomy Distribution across 133 Non-Equivalent Queries.

- 1. Schema Missing Join Path (37 queries, 27.8% of errors / 7.4% overall):** Omission of intermediate bridging tables (e.g., joining `order_reviews` to customers without traversing orders).
- 2. Semantic Filter Omission or Error (33 queries, 24.8% of errors / 6.6% overall):** Missing domain-specific predicates (e.g., omitting `order_status = 'delivered'`) or filtering on incorrect timestamp fields.
- 3. Semantic Aggregation Mismatch (32 queries, 24.1% of errors / 6.4% overall):** Discrepancies in aggregation function types (e.g., `AVG` vs `SUM` or unrounded currency amounts).
- 4. Schema Hallucinated Table (15 queries, 11.3% of errors / 3.0% overall):** Synthesizing nonexistent subqueries or table aliases not present in the catalog.
- 5. Grain / GROUP BY Mismatch (14 queries, 10.5% of errors / 2.8% overall):** Grouping by year-month string aliases rather than raw date expressions.
- 6. Semantic Ranking Order Mismatch (2 queries, 1.5% of errors / 0.4% overall):** Inverted or missing `ORDER BY` clauses in top-k rankings.

VII. ROBUSTNESS AND CROSS-SCHEMA TRANSFER

A. Controlled Synthetic Perturbation Robustness

To examine sensitivity to controlled lexical and semantic shifts, we evaluated a deterministic 50-query synthetic perturbation suite ($N = 50$, 10 queries per vector across 8 domains; Table IV and Fig. 7).

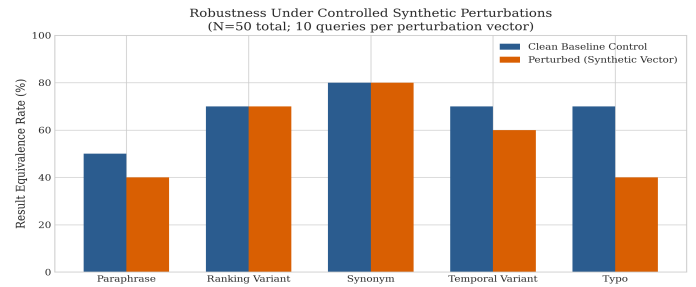


Fig. 7. Robustness Under Controlled Synthetic Perturbations ($N = 50$ total; 10 queries per perturbation vector).

TABLE IV
ROBUSTNESS UNDER 5 CONTROLLED PERTURBATION VECTORS ($N = 50$)

Perturbation Vector	Manipulation Description	Clean Acc	Perturbed Acc	Absolute Δ Acc	Retention Rate
Paraphrasing	Rephrasing query phrasing while preserving semantics	50.0%	40.0%	-10.0%	80.0%
Ranking Variants	Inverting top-k / bottom-k ordering phrasing	70.0%	70.0%	0.0%	100.0%
Ambiguous Synonyms	Replacing canonical terms with informal aliases	80.0%	80.0%	0.0%	100.0%
Temporal Shifts	Shifting date intervals and seasonal quarters	70.0%	60.0%	-10.0%	85.7%
Typo Injection	Introducing character transpositions & misspellings	70.0%	40.0%	-30.0%	57.1%

The pipeline demonstrates high retention under query paraphrasing (80.0% retention), ranking phrasing variants (100.0% retention), and synonym substitution (100.0% retention). Temporal boundary shifts exhibit moderate degradation (85.7% retention). However, character-level typographical noise causes severe degradation (57.1% retention, 30.0% absolute drop), exposing the vulnerability of exact token-matching in the schema retriever.

B. Cross-Schema Transfer Evaluation (Spider Probe)

To examine whether performance in a controlled, explicitly modeled relational warehouse transfers to heterogeneous unseen schemas, we evaluated the identical multi-stage pipeline on a stratified transfer probe of 50 queries across 20 distinct SQLite databases from the public Spider benchmark (Table V).

TABLE V
IN-DOMAIN VS. ZERO-SHOT CROSS-SCHEMA TRANSFER (SPIDER)

Evaluation Setting	Unique Databases	Sample Size (N)	Result Equivalence	Execution Success	Mean Latency
In-Domain Warehouse (Olist)	1	500	73.4%	100.0%	64.04s
Spider Transfer Probe	20	50	18.0%	100.0%	72.66s

Under zero-shot transfer, the system preserved execution reliability (100.0% SQL Execution Success across all 20 external databases) but experienced a substantial decline in semantic equivalence, achieving 18.0% Result Equivalence (9/50 matches). Inspection of generated SQL traces indicates that schema-linking assumptions tuned to the domain-specific foreign-key graph were a major source of transfer failures alongside unannotated cross-table joins. This empirical result demonstrates that strong performance within an explicitly modeled enterprise data warehouse does not automatically transfer to heterogeneous, uncurated external schemas without database-specific catalog introspection and schema tuning.

VIII. LIMITATIONS AND THREATS TO VALIDITY

- 1. Single Primary Relational Data Warehouse:** The primary 500-query benchmark is conducted on a multi-table relational e-commerce schema (Olist dataset, 9 tables, 100,000+ orders) with SQLite execution. Schema grounding assumptions tuned for this warehouse schema do not automatically translate to arbitrary domains.
- 2. Cross-Schema Transfer Gap (18.0% Spider Result):** On the 20-database, 50-query Spider transfer probe, result equivalence dropped to 18.0%, demonstrating a pronounced semantic-generalization gap on heterogeneous external schemas.
- 3. Sample Size of Transfer Probe:** The Spider transfer evaluation is a targeted 50-query stratified probe across 20 databases, not a full benchmark run.
- 4. Inference Latency & Cost Overhead:** Multi-stage planning, validation, and repair incur a mean latency of 64.04s (p_{95} : 121.92s) and higher token usage compared to single-shot prompting (~ 7 s).
- 5. Hard Query Complexity Ceiling:** Result equivalence drops to 44.55% on hard-tier queries and 32.56% on complex aggregated multi-column tables, reflecting persistent challenges in multi-step CTE nesting and window-function synthesis.
- 6. Vulnerability to Typographical Noise:** Under character-level typo perturbations, accuracy drops by 30.0% (57.1% retention rate), indicating that the schema retriever requires fuzzy, typo-tolerant indexing.
- 7. False-Positive Repair Regressions:** 21.8% of repair attempts degraded previously correct queries in this setting, confirming that syntactic repair success is not equivalent to semantic recovery.
- 8. Empirical Comparator Limits:** The row-multiset comparator is an empirical evaluation metric under the study comparator, not a formal proof of semantic correctness.
- 9. Sub-study Sample Sizes:** Component ablations ($N = 100$) and synthetic perturbation tests ($N = 50$) use smaller samples than the main 500-query benchmark.

IX. REPRODUCIBILITY

All code, benchmark definitions, evaluation scripts, and manuscript sources are open-source for full reproducibility:

- Complete reproduction instructions in `REPRODUCIBILITY.md`.
- Deterministic database constructor in `data/build_database.py`.
- Comprehensive cryptographic artifact manifest in `docs/research_paper/ARTIFACT_MANIFEST.json`.
- Verified MIT code license and Olist CC BY-NC-SA 4.0 data attribution in `LICENSE`.

X. CONCLUSION

In this study, we investigated the architectural mechanisms governing the reliability of LLM-generated analytical SQL over a public relational e-commerce data warehouse. On an audited 500-query benchmark, our multi-stage pipeline achieved a **73.40% Result Equivalence Rate under the study comparator** and **100.00% SQL Execution Success**. Our controlled component ablation demonstrated that adding the AST-based verification and repair stage provides a statistically significant improvement over unverified planning (Config C 26.0% vs. Config B 15.0%, exact $p = 0.0192$, $OR=3.75$). However, an exhaustive audit of 101 repair cases revealed that automated self-repair is a double-edged mechanism, producing **22 harmful false-positive regressions** against only **4 genuine recoveries**. Furthermore, the Spider transfer probe shows that these reliability gains are not automatically preserved under unseen schemas, with result equivalence falling to 18.0% despite maintaining 100.0% execution success. We conclude that conservative, uncertainty-aware structural verification is preferable to unconstrained automated repair loops.

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